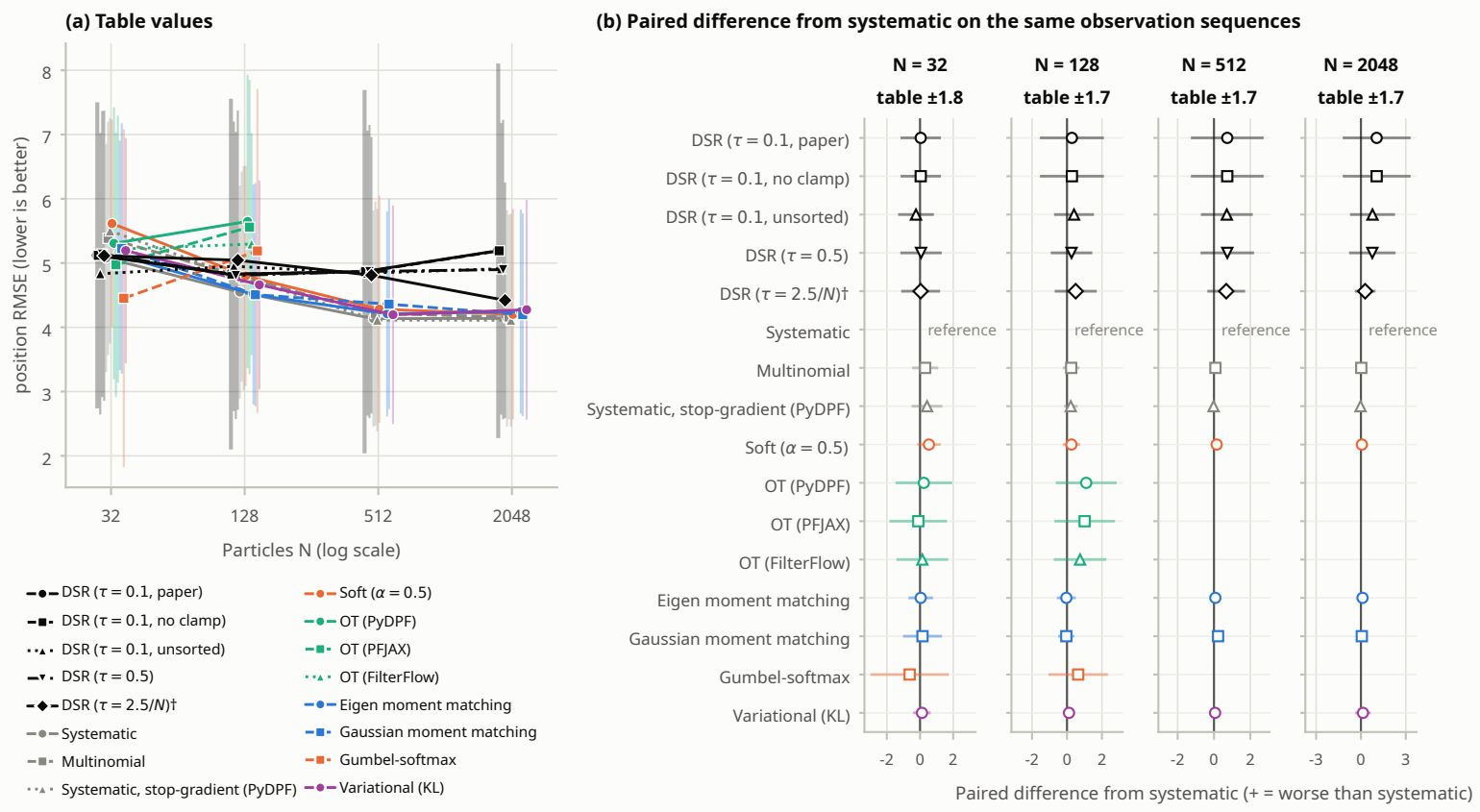

(a) Table values
position RMSE (lower is better)
8
7
6
5
4
3
2
32
128
512
2048
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
(b) Paired difference from systematic on the same observation sequences
N = 32
table ±1.8
N = 128
table ±1.7
N = 512
table ±1.7
N = 2048
table ±1.7
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
reference
-2
0
2
-3
3
Paired difference from systematic (+ = worse than systematic)